SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I K. YATHEESH (192412669) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 32 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO-1 (short)

Assessment Tool - 1 (19)

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CSA0708 - Computer Networks

7) Sol:-

Given

Total data Transferred = 3 GB

Effective throughput = 120 Mbps

Request size = 3 MB

32/40

i) The Application layer divides the file into smaller Requests of 3 MB each

$$\text{Number of Requests} = \frac{\text{Total data}}{\text{Request size}}$$

$$= \frac{3 \times 1024}{3}$$

$$= 1024 \text{ MB}$$

∴ Number of Requests = 1024 MB

ii) Calculate the transmission time

$$3 \times 1024 \times 1024 \times 8 = 25,165,824 \text{ bits}$$

$$\text{Time} = \frac{\text{Total data in bits}}{\text{Throughput}}$$

$$= \frac{3 \times 1024 \times 1024 \times 8}{120,000}$$

$$= \frac{25165824}{120000}$$

= 209.72 sec. //

∴ Total Requests = 1024

Transmission time = 209.72 sec.

* Application layer provides Reliable services:-

=> provides services such as FTP as for file transfer

=> Allow users to access network applications

=> Supports Authentication and access control

=> Detects application-level errors and handles failed requests.

=> works with the transport layers, which provide Reliable delivery using TCP when Required.

* Communication with the Transport layer:-

=> The Application layer uses transport services such as TCP to support reliable file delivery. TCP manages acknowledgments, retransmissions, & the correct ordering of data.

* Data Representation:-

=> Applications define how information such as files messages, and Records is interpreted and processed.

Conclusion:-

=> The system generates 1024 Requests od Requires approximately 209.72 sec to transmit 3 GB at 120 Mbps. the application layer provides file-transfer services. while Reliable delivery is supported by the underlying Transport layer when TCP is used.

Given

Application layer = 4 ms

Transport layer = 3 ms

Network layer = 5 ms

Data link layer = 2 ms

physical layer = 1 ms

propagation delay = 18 ms

Transmission delay = 12 ms

i) calculate Total processing dley.

4+3+5+2+1 = 15 ms

ii) calculate the total end-to-end delay

Total dlay = processing dulay + propagation dllay + Transmission dley.

$= 15 + 18 + 12$

$= 45$ ms

∴ Total end-to-end delay = 45 ms.

* Contribution of each OSI layer :-

=> Application layer :- provides network services to user applications

=> Transport layer :- Divides data into segments and manages end-to-end delivery.

=> Network layer :- Select Routes and forwards packets using IP addresses.

=> Data link layer :- creates frames and uses MAC addresses for local delivery.

=> Physical layer :- Transmits bits through cables or wireless signals.

$$D_{trans} = \frac{\text{Packet size in bits}}{\text{link Rate in bits/s}}$$

* Importance of end-to-end delay :-

=> low delay improves Real-time communication, video calls, and Online gaming.

applications

=> Healthcare monitoring & industrial IOT systems may require low latency for timely responses

=> Network engineers analyze processing propogation, transmission and queuing delays to identify performee bottlenecks

Conclusion :-

=> The total end-to-end delay is 45 ms each OSI layer perfoms a specific function, and it's processing time contributes to overall delay experinced by the packets..

Given

data = 80 mb

Size = 1600 bytes

frone bytes = 80 bytes

1 MB = 1,000,000 bytes.

i) calculate the useful data.

useful data per frame = frame size - Overhead

= 1600 - 80

= 1520 bytes.

ii) calculate the total no-of frames:-

80 MB = 80 × 1,000,000

= 80,000,000 bytes

$$N = \frac{\text{Total useful data}}{\text{useful data per frame}}$$

$$= \frac{80,000,000}{1520}$$

= 52631.58

∴ N = 52631.58 frames

iii) calculate the total protocol Overhead.

Total Overhead = N × Overhead

= 52632 × 80

= 4210560 bytes

$$\frac{4210560}{1000000} = 4.21 \text{ MB}$$

iv) calculate the total Transmitted data.

Total transmitted data = 52632 × 1600

= 84211200 byte

) efficiency

$$\text{efficiency} = \frac{80.000.000}{84.211,200} \times 100$$

$$= 94.998\ \%$$

$$= 95\ \%.$$

* **effects of Protocal Overhead:-**

=> Overhead increases the total amount of data transmitted.

=> It consumes additional bandwidth

=> It can increase transmission time.

1c)

Sol:- Given

Medical Record = 240 MB

compression = 30 %

segment size = 1400 bytes

Overhead = 60 bytes.

1 MB = 1,000,000 bytes

i) **calculate the compressed file:-**

Compressed size = $240 - (240 \times 0.30)$

$= 240 - 72$

$= 168$ MB.

ii) calculate the total no-of segments

$168 \times 1,000,000 = 168,000,000$

$= \frac{168,000,000}{1400}$

$= 120,000$ segments

iii) calculate the total protocol :-

Overhead $= 120,000 \times 60$

$= 7,200,000$ bytes

iv) calculate the final amount of data

Total transmitted data = compressed data + overhead

$= 168 + 7.2$

$= 175.2$ MB

* Multiple OSI layers :-

=> Application layer :- provides healthcare applications for sending and Receiving patient Records.

=> presentation layer :- Compresses the medical Record and may encrypt data to protect patient confidentiality.

=> Session layer:- Manages communication sessions b/w healthcare devices & servers.

=> Transport layer :- Divides data into segments and can use TCP acknowledments & Retransmissions for Reliable delivery.

=> Network layer:- Routes packets b/w healthcare devices and the hospital server.

=> Data link layer :- Add's frame headers and trailes and performs error detection.

=> Physical layer:- Transmits the data through wired (or) wireless communication media.